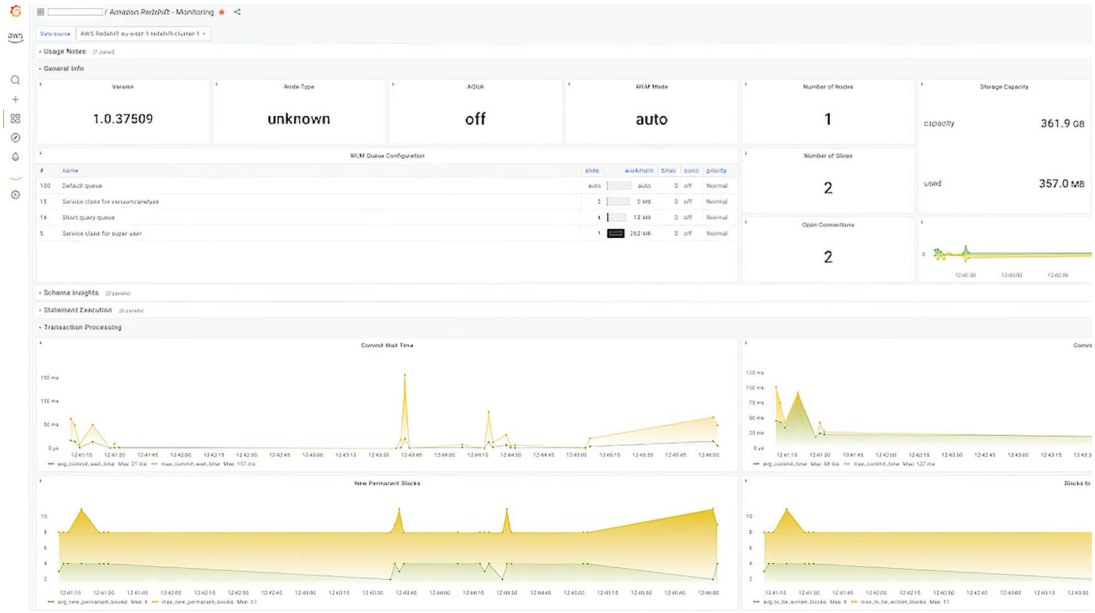


Figure 3.5 Amazon Redshift performance monitoring using CloudWatch as a data source in Grafana

With SQL, a declarative language, you state what the desired operations are, not how to execute them. The RDBMS executing your SQL query has a dedicated component called the *query planner* (https://sqlite.org/queryplanner.html), which takes care of optimizing the SQL query, taking indexes and other factors into account. You can learn about details of the query planning, for example, via the `EXPLAIN` statement in PostgreSQL (https://explain.depesz.com/s/Q6sQ).

NoSQL

NoSQL data stores have not been around as long as relational databases (less than 20 years compared to 50+ years in the case of RDBMSs). While, for logs, the same considerations apply as for RDBMSs: for performance metrics, we are faced with different characteristics. The emphasis in the context of NoSQL is less on query planning because the lack of relations means query planning matters less. Most important in the NoSQL context are access patterns and what is going on in the storage layer.

Given the breadth of NoSQL datastores (from key–value and document datastores, such as Redis, MongoDB, and DynamoDB, to graph datastores, like Neo4j or Amazon Neptune, to search engines, including Elasticsearch/OpenSearch), we won't go into great detail here, but a few examples may highlight good practices:

- "How to Monitor Redis with Prometheus" by David de Torres Huerta (https://sysdig.com/blog/redis-prometheus/)
- "Monitoring for MongoDB" (http://mng.bz/pPrK)